Certified Robustness for Model-Free PID Tuning by Step-Response Inspection

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Abstract

The SPIN method tunes a PID controller without a model. After each setpoint step, it counts how many turns the control error makes in three phase portraits, and it raises the gains until a count exceeds its limit. The limits were found by calibration, so robustness was not guaranteed. We prove two results. First, for any loop, the count of the integral portrait is within half a turn of half the number of setpoint crossings if the lobes of the error shrink. For loops that cancel the plant lags, the lobes shrink whenever $M_s < 26$, the sensitivity peak M_s grows strictly with the loop gain, and the error never crosses the setpoint if and only if $M_s \leq 1.394$. Second, for any stable digital loop with one degree of freedom, the increments of the error after a unit setpoint step form the impulse response of the sensitivity function. One finite record thus gives a lower and an upper bound on M_s that need no model, only a bound on the variation of the error after the record ends. As the acceptance test of SPIN, the upper bound certifies every accepted tuning, and the test never traps SPIN. Simulations on 19 loops confirm the theory.

1 Introduction

Let $L = CP$ be the loop transfer function and $S = 1/(1+L)$ the sensitivity function. Its peak $M_s = \max_\omega |S|$ measures robustness: a value near 1.4 is safe, and a value near 2 indicates poor robustness [2]. Tuning rules reach a chosen M_s by using a model [2, 3]. Once a loop is tuned, M_s is rarely checked, because a check needs either a model or a test that disturbs production [5, 6].

1.1 The SPIN method

SPIN [1] tunes a PI or PID controller with the gains K_i, K_p, K_d without a model. In each iteration, it applies one setpoint step, records the control error e, and forms three phase portraits,

$$\Gamma_0 : (e, \textstyle\sum e), \quad \Gamma_1 : (\Delta e, e), \quad \Gamma_2 : (\Delta^2 e, \Delta e), \tag{1}$$

where $\sum e$ is the running sum of e and Δ is the difference between consecutive samples. The portraits $\Gamma_0, \Gamma_1, \Gamma_2$ correspond to the frequency bands of K_i, K_p and K_d, which we call bands 0, 1 and 2. The turn count N_k is the number of turns that Γ_k makes around its end point before the error stays within $\pm\delta$ (2% of the step by default). Algorithm 1 states the rule. SPIN needs neither a model nor plant-specific settings [1]. However, its limits $\bar{N}_k$ were found by calibration, and the robustness of the result was not guaranteed. An online simulator lets the reader try SPIN.[1]

Algorithm 1: one iteration of SPIN [1]

```
apply a setpoint step and record the error e
if the record does not settle then
    divide K_i, K_p, K_d by 2, 4, 8                (retreat)
else
    count the turns N_0, N_1, N_2 of Γ_0, Γ_1, Γ_2
    if N_k ≤ N̄_k for every band k then
        raise all gains by one increment           (accept)
    else
        k ← lowest band with N_k > N̄_k
        lower the gain of band k by one increment and
        raise the gains of the bands below k by one
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One increment multiplies or divides a gain by $1/(1-\beta)$. Defaults: $\bar{N} = (0.5, 0.75, 1.0)$, $\delta = 0.02$, $\beta = 0.1$; each gain stays within $[0.001, 10]$ times its initial value (the gain box).

1.2 Contributions

The paper answers three questions, each building on the previous one.

1. *What does a count measure?* (Section 2) A portrait turns by exactly half a turn between two consecutive zeros of its first coordinate, in any linear coordinates (Lemma 1), so the count of Γ_0 is within half a turn of half the number of setpoint crossings (Corollary 1). On loops that cancel the plant lags, the crossings reveal M_s: the error never crosses the setpoint if and only if $M_s \leq 1.394$ (Propositions 1, 2). Elsewhere, counts and M_s are not tied.

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[1] https://robopid-simulator.uceeb.cvut.cz

2. *Can the step show M_s anyway?* (Section 3) Yes. For a digital loop, the record gives the sensitivity function exactly (Lemma 2), and one finite record gives certified lower and upper bounds on M_s (Theorem 1).
3. *Can SPIN then guarantee robustness?* (Section 4) Yes. With the upper bound as its acceptance test, SPIN accepts only tunings with $M_s \le M^*$, the test never traps it (Theorem 2), and the same record predicts which gain to lower (Proposition 3).

Section 5 confirms every statement in simulation.

Related work Frequency responses, sensitivity functions and margins have been estimated from step or relay responses and from closed-loop experiments [7, 8, 9]; PI controllers have been tuned from one setpoint step [14]; and model-free methods optimize performance from repeated experiments or one data set [12, 13, 15]. Here, one ordinary setpoint step certifies M_s inside a model-free tuning rule.

2 Counts Count Crossings

2.1 Every crossing adds half a turn

In each portrait (1), the second coordinate q is the running sum of the first coordinate p; in continuous time, $\dot{q} = p$. The turns are counted around the end point, which we place at the origin. We state the results in continuous time; the sampled counts of SPIN approximate them. For Γ_0, the zeros of $p = e$ are the setpoint crossings; for Γ_1, they are the extrema of e.

Lemma 1 (Half turns) *Let $\dot{q} = p$ with p continuous, and let $t_1 < t_2$ be consecutive zeros of p with $q(t_1)q(t_2) < 0$. Then, on $[t_1, t_2]$, the curve (p, q) turns counterclockwise around the origin by exactly half a turn. The same holds for its image under any linear map with a positive determinant, such as the whitening used in [1].*

Proof: On (t_1, t_2), p keeps one sign. Hence the curve stays in one open half-plane bounded by the q-axis, and q moves monotonically from $q(t_1)$ to $q(t_2)$, that is, from one ray of the q-axis to the opposite ray. Within the half-plane, the angle stays in an open interval of length π, so it changes by exactly π. If $p > 0$, q increases; if $p < 0$, q decreases; in both cases the turn is counterclockwise. A linear map with a positive determinant maps the half-plane to a half-plane and the two rays to opposite rays, and it preserves the orientation. □

Corollary 1 (Counts and crossings) *Let p change sign at $t_1 < \cdots < t_m$, $m \ge 1$, before the error stays within $\pm\delta$. Let $q(t_j)$ alternate in sign, and let $q(t_1)$ have the sign of p before t_1. Then the turn count of (p, q) lies in $(\frac{m-1}{2}, \frac{m+1}{2})$. For Γ_0, these conditions hold if the areas of the lobes of e between crossings decrease strictly.*

Proof: By Lemma 1, each of the $m-1$ intervals between sign changes adds π. Before t_1 and after t_m, the curve stays in one open half-plane and starts or ends on the ray from which it turns counterclockwise; so each end adds an angle in $(0, \pi)$. For Γ_0, $q(t_j) = -\int_{t_j}^{\infty} e\,dt$ is an alternating sum of lobe areas; if the areas decrease, the sum has the sign of its first term. □

2.2 When crossings reveal M_s

Lambda, IMC and SIMC tunings cancel the plant lags [3]. Examples are a PI controller $K_c(1 + 1/(T_i s))$ with $T_i = T$ on the plant $Ke^{-sL_d}/(Ts+1)$, or a series PID controller with $T_i = T_1$ and $T_d = T_2$ on a plant with two lags. The loop is then

$$L(s) = \frac{k\,e^{-sL_d}}{sL_d}, \qquad k = \frac{KK_cL_d}{T_i}. \tag{2}$$

Proposition 1 (Exact robustness) *The loop (2) is stable for $0 < k < \pi/2$. Its gain margin is $\pi/(2k)$, and its poles are $W_j(-k)/L_d$, where W_j are the branches of the Lambert W function [10]. Its peak M_s is a strictly increasing function of k, given by*

$$k = x\sin x - x^2\cos x, \qquad \frac{1}{M_s^2} = 1 - \frac{2k\sin x}{x} + \frac{k^2}{x^2}, \tag{3}$$

where $x = \omega_p L_d \in (0, \pi/2)$ is the normalized peak frequency.

Proof: The phase of L is $-\pi/2 - x$. It reaches $-\pi$ at $x = \pi/2$, where $|L| = 2k/\pi$. The poles solve $sL_d e^{sL_d} = -k$. Let $h(x) = |1+L|^2 = 1 - 2k\sin x/x + k^2/x^2$. Then $h'(x) = 2k(\varphi(x) - k)/x^3$, where $\varphi(x) = x\sin x - x^2\cos x$ increases on $(0, \pi)$ and $\varphi(\pi/2) = \pi/2 > k$. Hence, on $(0, \pi)$, h has a single minimum $x^* \in (0, \pi/2)$. On $[\pi, 2\pi]$, $h \ge 1$; for $x \ge 2\pi$, $h \ge (1 - k/x)^2 > (1 - 2k/\pi)^2 = h(\pi/2)$. So x^* is the global minimum, which gives (3). By the envelope theorem, $\frac{d}{dk}h(x^*) = \frac{2}{x^{*2}}(k - x^*\sin x^*) = -2\cos x^* < 0$. □

SIMC with $\tau_c = L_d$ and $T \le 8L_d$ cancels the lag and gives $k = \frac{1}{2}$. Hence the gain margin is π and, by (3), $M_s = 1.59$, which are exactly the values in [4, Table 2].

After a unit setpoint step, $e = 1$ on $[0, L_d]$ and $\dot{e}(t) = -(k/L_d)\,e(t - L_d)$. Let $c_1 < c_2 < \ldots$ be the setpoint crossings, let $c_0 = 0$, let lobe i be the interval $[c_i, c_{i+1}]$, and let $E_0 = 1$.

Proposition 2 (Crossings) *For the loop (2) with $0 < k < \pi/2$, the following hold.*

(a) The error never crosses the setpoint if and only if $k \le 1/e$, that is, if and only if $M_s \le M_s(1/e) = 1.394$. For $k > 1/e$, it crosses infinitely often.

(b) Lobe i contains exactly one extremum E_i, at $c_i + L_d$.

(c) $|E_{i+1}| \le r(k)\,|E_i|$, where $r(k) = k^2/2$ for $k \le 1$ and $r(k) = k - \frac{1}{2}$ for $k \ge 1$.

(d) For $k < 3/2$, the lobe areas decrease strictly, so Corollary 1 holds for Γ_0: N_0 is within half a turn of half the number of crossings.

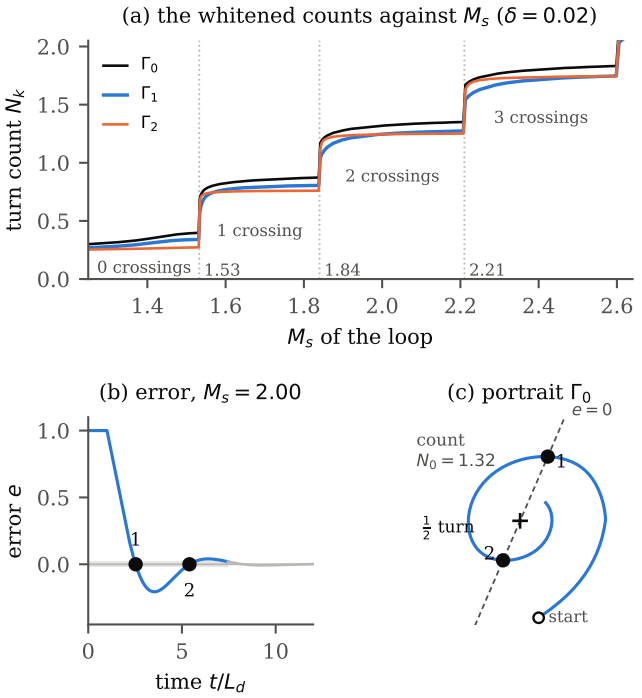


Figure 1: The lag-cancelled loop. (a) Whitened counts versus M_s ($\delta = 0.02$); N_0 stays within half a turn of half the number of crossings (Corollary 1). (b) A record with $M_s = 2.00$ and two crossings. (c) Its portrait Γ_0 turns by exactly half a turn between the crossings (Lemma 1); the radius is compressed but the angle is not.

Proof: (a) The record is the fundamental solution of the delay equation. It is positive if and only if $s + (k/L_d)e^{-sL_d} = 0$ has a real root, that is, if and only if $k \le 1/e$; for $k > 1/e$, all solutions oscillate [11]. The value of M_s follows from Proposition 1. (b) The sign of $\dot{e}(t)$ is that of $-e(t-L_d)$. By induction over the lobes, e moves away from zero on $(c_i, c_i + L_d)$, where $e(t-L_d)$ lies in lobe $i-1$; hence lobe i is longer than L_d. On $(c_i + L_d, c_{i+1} + L_d)$, e moves toward zero. For $i = 0$, the extremum is the plateau $e = 1$ on $[0, L_d]$. (c) On lobe i, $|e| \le |E_i|$. After the extremum, $|\dot{e}| \le (k/L_d)|E_i|$, so e needs at least L_d/k to reach zero, and $|e(t)| \le |E_i| \min\{1, k(c_{i+1} - t)/L_d\}$ on the last L_d of the lobe. Since $e(c_{i+1}) = 0$, $E_{i+1} = -\frac{k}{L_d}\int_{c_{i+1}-L_d}^{c_{i+1}} e\,dt$, and integrating the bound gives $r(k)|E_i|$. (d) Integrating $\dot{e}$ from $c_i + L_d$ to $c_{i+1} + L_d$ gives the area $\frac{L_d}{k}(|E_i| + |E_{i+1}|)$ of lobe i, because the extrema alternate in sign. By (c), $|E_{i+2}| < |E_i|$ if $r(k) < 1$, that is, if $k < 3/2$, where $M_s = 26.2$. □

Thus N_0 reveals the number of crossings, and the crossings reveal M_s (Fig. 1).

2.3 When they do not

Outside lag-cancelled loops, crossings and M_s are not tied. Consider a first-order plant with $T = 0.3L_d$, controlled by PI with $T_i = 8T$ and a gain margin of 1.624. Its M_s is 2.62, yet its error never crosses the setpoint by more than 2%, because the oscillation rides on a slow decay. (This is the M_s of the continuous loop; the digital loop of Fig. 2 has 2.61.) Conversely, overshoot caused by a controller zero can produce crossings at a low M_s. Counts can therefore guide tuning, but they cannot certify it. The same step, however, contains M_s.

3 One Step Reveals M_s

A digital controller runs with the sampling period T_s. Seen at the samples, the loop it closes is linear and time invariant. From here on, $L(z)$, $P(z)$, $S = 1/(1+L)$ and $M_s = \max_{\theta\in[0,\pi]} |S(e^{j\theta})|$, with $\theta = \omega T_s$, refer to this digital loop, the loop that actually runs.

Lemma 2 (Exact identity) *Let the loop be stable, have one degree of freedom, and have a strictly proper L (the measurement precedes the control action). Let e_n be the error after a unit setpoint step at $n = 0$, applied from rest and without disturbances, so that $e_{-1} = 0$ and $e_0 = 1$, and let $a_n = \Delta e_n = e_n - e_{n-1}$. Then*

$$S(e^{j\theta}) = \sum_{n\ge 0} a_n\, e^{-jn\theta}. \tag{4}$$

Proof: $E(z) = S(z)R(z)$ with $R(z) = 1/(1-z^{-1})$, so $S = (1-z^{-1})E$. In a stable loop, a_n decays exponentially, so the series converges on the unit circle. □

A real record is finite, and the only quantity it cannot provide is the remaining variation of the error after its end.

Assumption 1 (Long enough record) *The loop is stable, and the record $e_0, \dots, e_N$ satisfies $\sum_{n>N} |a_n| \le \tau$ for a known τ.*

In a stable loop, $|a_n| \le c\rho^n$ with $\rho < 1$, so the remaining variation falls geometrically with the record length.

Theorem 1 (Certified bounds from one record) *Let Assumption 1 hold, and let $\hat{S}(\theta) = \sum_{n=0}^{N} a_n e^{-jn\theta}$. Let G be a uniform grid on $[0, \bar{\theta}]$, $\bar{\theta} \le \pi$, that has spacing h and contains both ends. For any n_c, let*

$$J = \textstyle\sum_{n=0}^{N} |n - n_c|\,|a_n|, \quad B = \sum_{n=1}^{N-1} |a_{n+1} - a_n| + |a_N| + 2\tau.$$

Then $\underline{M} \le M_s \le \overline{M}$, where

$$\underline{M} = \max_{\theta\in G} |\hat{S}(\theta)| - \tau,$$
$$\overline{M} = \max\left\{ \max_{\theta\in G} |\hat{S}(\theta)| + \tfrac{hJ}{2} + \tau,\ 1 + \tfrac{B}{\sin(\bar{\theta}/2)} \right\}, \tag{5}$$

and the second term is omitted if $\bar{\theta} = \pi$. If τ equals the true remaining variation, $N \to \infty$, $h \to 0$ and $\bar{\theta} = \pi$, then both bounds converge to M_s.

Proof: By (4), $|S - \hat{S}| \le \sum_{n>N} |a_n| \le \tau$ at every θ, which gives $\underline{M}$. On $[0, \bar{\theta}]$, $|\hat{S}|$ equals $|e^{jn_c\theta}\hat{S}|$, whose derivative is bounded

Algorithm 2: certified SPIN; line 7 of Algorithm 1 is replaced by

7a: compute $\overline{M}$ by Theorem 1, with $\bar{\theta}$
from $1+B/\sin(\bar{\theta}/2)=M^*$ ($\bar{\theta}=\pi$ if no solution)
and the grid spacing h such that $hJ/2\le\eta$
7b: **if** $\overline{M}\le M^*$ **then** raise all gains by one increment (accept)
7c: **else** predict by Proposition 3 the M_s after
each band move (lower one band, raise those below)
7d: make the band move of the highest band predicted
to lower M_s, at most $n_{\max}$ times between
acceptances; otherwise lower all gains by one increment

Defaults: $M^*=1.7$, $\tau=10^{-3}$, $\eta=2\cdot10^{-3}$, $n_{\max}=10$.

by J. Every θ lies within $h/2$ of a grid point, which gives the first term of $\overline{M}$. On $[\bar{\theta},\pi]$, let $z=e^{-j\theta}$ and $Z_n=\sum_{m=1}^{n}z^m$, so that $|Z_n|\le 2/|1-z|=1/\sin(\theta/2)$. Summation by parts gives $\sum_{n\ge1}a_nz^n=\sum_{n\ge1}(a_n-a_{n+1})Z_n$. Since $\sum_{n\ge N}|a_n-a_{n+1}|\le|a_N|+2\tau$, we obtain $|S-a_0|\le B/\sin(\theta/2)$ with $a_0=1$. Convergence follows because the remaining variation and hJ tend to zero. □

Every quantity in (5) except τ is computed from the record, and all grid values follow from one chirp z-transform. The two terms of $\overline{M}$ split the frequency axis: (1) on $[0,\bar{\theta}]$, the grid term bounds $|S|$ by its grid values; (2) on $[\bar{\theta},\pi]$, the second term needs no grid: for a smooth record, B is small and bounds all high frequencies at once.

4 SPIN That Never Accepts a Fragile Tuning

Algorithm 2 changes only line 7 of SPIN. The counts still shape the tuning, and the bound decides whether a tuning is accepted. After a rejection, it uses two kinds of moves: (1) the band move of line 10 of Algorithm 1, and (2) the uniform back-off, the reverse of line 7.

Theorem 2 (Certified SPIN) *Let the plant be stable, let $K_iP(1)>0$, let Assumption 1 hold for every record, and let $hJ/2\le\eta$. For (iii) and (iv), also let no gain be clipped by the gain box.*

(i) Every tuning accepted by Algorithm 2 has $M_s\le M^*$.

(ii) For gains K, let $m=\min\{0,\inf_{0<\theta\le\pi}\mathrm{Re}\,L(e^{j\theta})\}$. For $0<\lambda<1/|m|$, the loop with the gains λK is stable, and

$$M_s(\lambda K)\le\frac{1}{1-\lambda|m|}. \tag{6}$$

(iii) If $M^*>1+2\tau+\eta$, every run of consecutive rejections by the test of line 7b ends, either in an acceptance or in a move of Algorithm 1 (line 3 or line 10). The run contains at most $n_{\max}$ band moves, and each stretch of uniform back-offs in it has at most $\lceil\ln\frac{|m|M'}{M'-1}/\ln\frac{1}{1-\beta}\rceil$ of them, where $M'=M^*-2\tau-\eta$ and m belongs to the gains at the start of the stretch.

(iv) Every move except an acceptance lowers $\Phi=\ln K_i+2\ln K_p+4\ln K_d$ by at least $\ln\frac{1}{1-\beta}$. Hence the gains never return to earlier values without an acceptance in between.

Proof: (i) This follows from Theorem 1. (ii) L is bounded on the unit circle except near $z=1$. There, the integral term has a bounded real part, because P is analytic at $z=1$ and $\mathrm{Re}\,1/(e^{j\theta}-1)=-\frac{1}{2}$; so m is finite. Then $|1+\lambda L|\ge 1+\lambda\,\mathrm{Re}\,L\ge 1-\lambda|m|>0$, which gives (6). The Nyquist curve of λL lies in $\mathrm{Re}\ge-\lambda|m|>-1$, also on the indentation at $z=1$, where $K_iP(1)>0$ makes the real part large and positive. It therefore cannot encircle -1, and since L has no poles outside the unit disc, the loop is stable. (iii) Line 7d allows at most $n_{\max}$ band moves. In a stretch of uniform back-offs from gains K, $\lambda=(1-\beta)^j$ after j back-offs, and (6) gives $M_s\le M'$ for the stated j. In (5), the grid term is then at most $M_s+\tau+\eta+\tau\le M^*$, and the second term is at most M^* by the choice of $\bar{\theta}$; so the test passes unless the counts reject first. (iv) With $w=(1,2,4)$, a band move changes Φ by $\ln\frac{1}{1-\beta}(\sum_{j<k}w_j-w_k)\le-\ln\frac{1}{1-\beta}$. A uniform back-off changes it by $-7\ln\frac{1}{1-\beta}$, and line 3 by $-17\ln2$. Returning to the same gains requires $\sum\Delta\Phi=0$, hence an acceptance, which raises Φ. □

The test therefore never traps SPIN: it either certifies a tuning or returns control to the original rules. Part (i) holds with or without the gain box. Part (iv) does not exclude a drift without acceptance, in which Φ decreases forever. Such a drift is a property of the counts, and the gain box of SPIN stops it (Section 5).

A uniform back-off is safe by (ii), but it worsens load rejection. A band move is often cheaper, and the same record predicts what each move does.

Proposition 3 (Prediction from one step) *For any controller C' on the same plant, $S'=S/(S+(1-S)C'/C)$ exactly. If the record gives $\hat{S}=S+\varepsilon$, the predicted $\hat{S}'$ has the error $\hat{S}'-S'=\varepsilon\,(C'/C)\,S'\hat{S}'/(S\hat{S})$.*

Proof: $L'=LC'/C$ with $L=(1-S)/S$; subtracting the two expressions gives the error. □

The prediction only selects the move; the next iteration tests it, so safety never depends on the prediction.

5 The Theory at Work

Setup We compare SPIN as published with certified SPIN (Algorithm 2). Both run 200 iterations with a derivative filter ratio of 10 and the gain box of [1]. Because the box ties the result to the initial tuning, every loop runs from two initial tunings: SIMC on a half-rule model with τ_c equal to three times its dead time (cautious start), and the same tuning with all gains doubled (bold start). The test set has 19 loops.

1. The four plants of [1], with PI and with PID,

$$P_1 = \frac{e^{-s}}{(10s+1)(s+1)^3}, \qquad P_2 = \frac{1.25\,e^{-8s}}{(5s+1)^4},$$
$$P_3 = \frac{e^{-10s}}{(2s+1)(s+1)^3}, \qquad P_4 = \frac{e^{-4s}}{(8s+1)^6}.$$

2. Four plants that place the oscillation in a chosen band: the delay-dominant $e^{-5s}/(s+1)$ and the lag-dominant $e^{-s}/(10s+1)$ and $e^{-0.5s}/((10s+1)(s+1))$ with PI; the fast $e^{-0.1s}/((s+1)(0.5s+1))$ with PID.
3. The family $e^{-L_d s}/((s+1)(0.5s+1)(0.25s+1))$ with PID and $L_d \in \{0.25, 0.5, 1, \ldots, 16\}$, whose dead-time share runs from 0.125 to 0.90.

P, I and D all act on the error (one degree of freedom). Let T_a be the dead time plus the lag time constants of the plant and T_f the derivative filter time constant; the controller runs at $T_s = \min(T_a/100, T_f/20)$ for PID and $T_a/1000$ for PI, with exactly discretized lags and the delay rounded to whole samples. Each record covers $15\,T_a$. All values of M_s are those of this digital loop, computed from its exact L. Because SPIN ends in a cycle between a few tunings, we report the most aggressive tuning accepted in the last 60 iterations.

Counts (Section 2) On the lag-cancelled loop with $1/e < k < 3/2$, for all 126 values of k tested, (1) bound (c) of Proposition 2 held for every extremum, with equality for the first one when $k \geq 1$; (2) the lobe areas decreased, and Γ_0 turned by half a turn between crossings to within 10^{-6}, also after a linear map; (3) the whitened count of SPIN satisfied Corollary 1; (4) (3) matched a direct search for M_s to six digits.

Bounds (Section 3) For all 146 tunings accepted in the runs below, a record four times longer showed the true remaining variation $\sum_{n>N} |a_n| \leq 2.8 \cdot 10^{-5}$, so Assumption 1 held with $\tau = 10^{-3}$. The identity (4) held up to this variation, and every true M_s lay within its bounds, with $\overline{M} - M_s \leq \eta + \tau + 2.8 \cdot 10^{-5}$. On short records the bounds are wide, yet they contain M_s and tighten as the record grows (Fig. 2(b)).

Certified SPIN (Section 4) See Table 1 and Fig. 3.

- *Safety.* Published SPIN ended above $M^* = 1.7$ on 9 of the 19 loops from the cautious start (up to 2.01) and on 8 from the bold start (up to 2.09); which loops failed depended on the start, because the gain box often set the result. Certified SPIN ended at or below M^* on all loops from both starts (at most 1.69), as Theorem 2(i) guarantees.
- *Without the gain box.* Published SPIN exceeded M^* on 12 of the 18 loops that accepted a tuning in their last 60 iterations (up to 2.05), whereas certified SPIN stayed at or below M^* on all 18 (up to 1.68).
- *Back-off and prediction.* Along the uniform back-off from the most aggressive published tuning of each loop (cautious start), the loop stayed stable and (6) held. The predicted M_s after every band move matched the true value to within 10^{-7} (Fig. 3(b)).

Table 1: M_s of the digital loop for the most aggressive tuning that published and certified SPIN accept, from the cautious and the bold start, and the change of the load-step IAE (cautious start; –: tuning unchanged).

		published		certified		
		caut.	bold	caut.	bold	ΔIAE
P_1	PI	1.46	1.46	1.46	1.46	–
	PID	1.66	**1.84**	1.66	1.61	–
P_2	PI	1.64	1.67	1.64	1.67	–
	PID	1.59	**1.78**	1.59	1.66	–
P_3	PI	1.67	1.63	1.67	1.63	–
	PID	**1.81**	**1.74**	1.69	1.60	+23%
P_4	PI	1.55	1.60	1.55	1.60	–
	PID	1.61	1.55	1.61	1.55	–
delay-dom.	PI	1.68	1.63	1.68	1.63	–
lag-dom.	PI	**2.01**	1.69	1.67	1.69	+52%
2 lags	PI	1.44	1.47	1.44	1.47	–
fast	PID	**1.89**	**1.98**	1.66	1.69	0%

- *No trapping.* Runs of rejections by the test contained at most 4 band moves, fewer than $n_{\max}$, and every loop had at least 7 acceptances in its last 60 iterations. The one exception is $L_d = 16$ without the box, which accepts no tuning under either rule: the counts reject in a repeating pattern that lowers K_d in every round, the drift that Theorem 2(iv) allows.
- *Cost.* The price is worse load rejection: the integrated absolute error (IAE) after a unit load step at the plant input rose by 7.8% on average over the 19 loops (6.4% from the bold start).

6 Conclusion

The three questions of the introduction have clear answers.

1. A count measures setpoint crossings; on lag-cancelled loops, an error that never crosses the setpoint means $M_s \leq 1.394$.
2. The setpoint step that SPIN already records gives the sensitivity function of a digital loop exactly, and one finite record certifies M_s without a model; so does every logged setpoint change made from rest.
3. With this certificate as its acceptance test, SPIN accepts only tunings with $M_s \leq M^*$, and the test never traps it.

The theorems assume a linear loop and a long enough record; noise and real loops are the subject of a follow-up. Scripts for every number and figure are available at `https://github.com/ottapav/robopid-simulator`, folder `docs/ECC27_certified_SPIN`.

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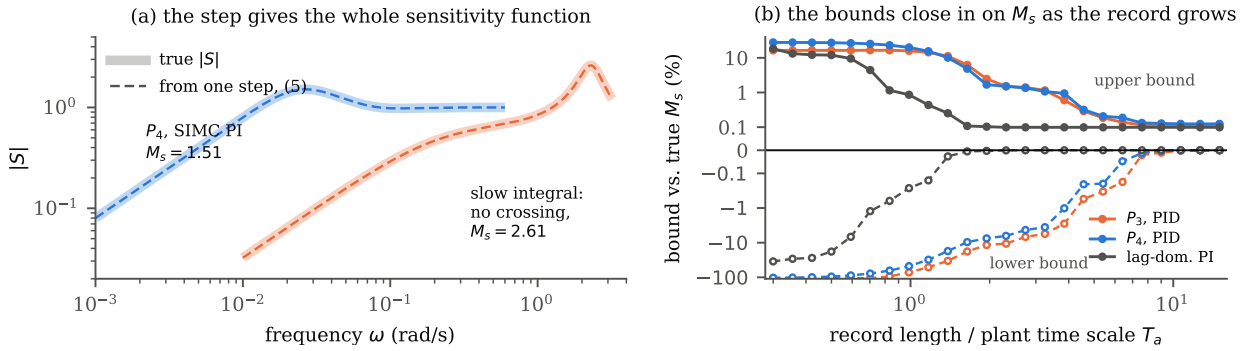


Figure 2: (a) $|S|$ from one setpoint step (dashed) lies on the true $|S|$ (thick), also for the slow-integral loop whose peak the counts miss (Lemma 2). (b) The bounds (5) relative to the true M_s versus the record length, with τ the true remaining variation (Theorem 1).

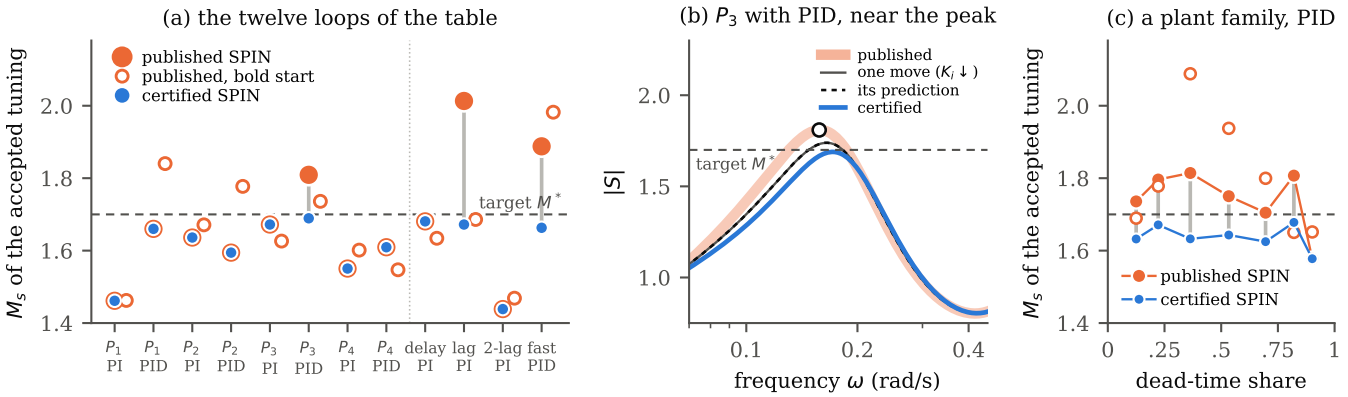


Figure 3: Published versus certified SPIN ($M^* = 1.7$). (a) The loops of Table 1. (b) P_3 with PID: published tuning ($\overline{M} = 1.81$, ○); one band move lowering K_i ($M_s = 1.739$, as predicted by Proposition 3); certified tuning (1.689). (c) The plant family versus its dead-time share.

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